ON LOGARITHMS

• The logarithm of a number x in base b is the number such that $x = b^n$,

e.g., $\log_3(81) = 4$ since $3^4 = 81$

e.g., $\log_{10}(100) = 2$ since $10^2 = 100$

• The most widely used bases, b, are

10,

2, and

e ≈ 2.71828..., defined because $d\log_e(x)/dx = 1/x$; "$\log_e$" is usually written as "ln"

• To convert from base k to base b use

$\log_b(x) = \log_k(x) / \log_k(b)$

NB. ln (10) = 2.302585....]

• The impact made by logarithms when they were discovered in the 17th century was similar to the impact made by the invention of the computer − they simplify calculations and so made it possible to attempt analyses that involved many more complex calculations than had been practical before. Thus multiplications are reduced to additions, divisions to subtractions, and raising a number to a power to a multiplication, all because

$\log(mn) = \log(m) + \log(n)$

$\log(m/n) = \log(m) - \log(n)$

$\log(m^n) = n \log(m)$

• For example, 2532.174 × 86491.221 can be determined by looking up the log of 2532.174 and the log of 86491.221, adding the two together, and then looking up the antilog. Special booklets of tables were created to allow one to look up the log and antilog of any number. Just 35 years ago these tables used to be a commonplace part of the schoolboy's kit.

• A particularly useful example of using logarithms comes with the following equation: if

$$\dot{\varepsilon} = A\sigma^n \exp(-H/RT)$$

then

$$\ln\dot{\varepsilon} = \ln A + n\ln\sigma - H/RT$$

and so

$$n = (\ln\dot{\varepsilon} - \ln A + H/RT) / \ln\sigma$$